SIREN: Stochastic Itō Residual on Embedded Networks

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Abstract

Network intrusion detection systems (NIDS) are increasingly adopting Graph Neural Networks (GNNs) to capture complex topological behaviours such as lateral movement. However, existing GNN-based approaches largely treat intrusion detection as a discrete-time, supervised classification task, making them vulnerable to zero-day exploits and incapable of natively modelling the continuous temporal dynamics of network traffic. In this paper, we propose SIREN (Stochastic Itō Residual on Embedded Networks), an unsupervised anomaly detection framework that models a monitored network as a continuous-time dynamical system. SIREN postulates that under normal operation, host traffic features evolve according to a system of graph-coupled Itō stochastic differential equations (SDEs), converging to a unique stationary distribution. We leverage score matching to learn the score function of this distribution from normal traffic data alone. During inference, SIREN continuously monitors the network state and computes the *Stein residual*—the discrepancy between the model-predicted score and the empirical score estimated via short-window Kernel Density Estimation (KDE). By dynamically aggregating these Stein residuals over the flow graph, SIREN propagates anomaly signals across connected neighbours, effectively exposing stealthy, coordinated lateral movement attacks. Extensive experiments on standard benchmark datasets, including UNSW-NB15 and CIC-IDS2018, demonstrate that SIREN achieves state-of-the-art detection performance while maintaining a significantly lower False Alarm Rate (FAR) compared to recent discrete-time GNN baselines.

Keywords: Intrusion Detection, Graph Neural Networks, Stochastic Differential Equations, Score Matching, Anomaly Detection

1 Introduction

The proliferation of Internet of Things (IoT) devices and the increasing complexity of enterprise networks have fundamentally shifted the landscape of cybersecurity. Modern cyberattacks, particularly Advanced Persistent Threats (APTs) and lateral movement strategies, no longer rely on isolated, easily detectable exploits. Instead, they manifest as coordinated sequences of subtle, individually stealthy actions distributed across multiple hosts [1]. Traditional Intrusion Detection Systems (IDS), which evaluate network flows or packets in isolation, frequently fail to capture the broader spatial and temporal context necessary to detect these sophisticated campaigns.

To address this limitation, recent research has increasingly turned to Graph Neural Networks (GNNs) for intrusion detection [2, 3]. By representing network entities (such as IP addresses or endpoints) as nodes and communication flows as edges, GNN-based IDSs explicitly model the topological structure of the network. Approaches like E-GraphSAGE [2] and recent multi-granularity frameworks [4] have demonstrated significant improvements in detecting complex attacks by leveraging message-passing mechanisms to aggregate neighbourhood information. However, the majority of existing GNN-IDS methods formulate the problem as a discrete-time, supervised classification task. This formulation suffers from two critical drawbacks: first, it requires extensive labelled data encompassing all possible attack vectors, leaving the system vulnerable to zero-day exploits; second, it treats network traffic as a sequence of static snapshots, failing to natively capture the continuous, underlying dynamical processes that govern network behaviour over time.

In this paper, we introduce SIREN (Stochastic Itō Residual on Embedded Networks), a fundamentally novel approach that models a monitored network as a continuous-time dynamical system on a graph. Rather than training a classifier to distinguish between specific attack signatures and benign traffic, SIREN learns the natural, continuous evolution of the network's state. We hypothesise that under normal operation, the network traffic features of interconnected hosts evolve according to a system of graph-coupled Itō stochastic differential equations (SDEs), eventually converging to a well-defined, unique stationary distribution.

SIREN leverages score-matching techniques [5] to learn the score function (the gradient of the log-probability density) of this normal stationary distribution. At inference time, the system continuously monitors the network state and computes the empirical score of the observed traffic. By measuring the discrepancy between the model-predicted score and the empirical score—a metric we term the *Stein residual*—SIREN can identify when a node's trajectory deviates from expected normal behaviour. Crucially, by aggregating these Stein residuals across the network graph, SIREN amplifies the weak anomaly signals produced by lateral movement attacks, exposing coordinated malicious activities that would remain hidden if hosts were analysed independently.

The main contributions of this work are as follows:

- We propose a continuous-time formulation of network intrusion detection, modelling host feature trajectories via graph-coupled Itō SDEs that capture both autonomous node dynamics and topological mean-reversion.

- We introduce the *Stein residual* as a mathematically grounded, unsupervised anomaly signal. By evaluating the discrepancy between a learned score network and short-window Kernel Density Estimates (KDE), our approach detects distributional shifts without relying on labelled attack data.
- We present a graph-aware aggregation mechanism for Stein residuals, specifically designed to propagate anomaly signals across connected hosts and expose coordinated lateral movement attacks.
- We conduct extensive experiments on standard benchmark datasets (UNSW-NB15 and CIC-IDS2018), demonstrating that SIREN outperforms state-of-the-art discrete-time GNN baselines, particularly in terms of False Alarm Rate (FAR) and robustness to feature noise and edge perturbations.

The remainder of this paper is structured as follows: Section 2 reviews related work in GNN-based intrusion detection and continuous-time graph models. Section 3 details the SIREN methodology, including graph construction, the SDE formulation, the score network architecture, and the training and inference algorithms. Section **??** presents our experimental setup and results, and Section **??** concludes the paper.

2 Related Work

2.1 Graph Neural Networks for Network Intrusion Detection

Graph neural networks (GNNs) have become the dominant paradigm for network intrusion detection owing to their capacity to exploit the structural relationships among hosts and traffic flows that elude flat feature-vector classifiers [6]. Lo et al. [2] proposed E-GraphSAGE, converting per-flow records into a network graph and applying GraphSAGE-based edge classification; despite its widespread use as a baseline, E-GraphSAGE operates on static snapshot graphs and emits uncalibrated point predictions. More recent work has sought to enrich both graph construction and representation learning. Ngo et al. [7] introduced TKSGF, replacing physical-link graphs with Top-K attribute-similarity edges paired with a GraphSAGE backbone, and demonstrated that feature-driven topology substantially improves detection accuracy on IoT benchmarks. Lin et al. [8] proposed E-GRACL, augmenting GraphSAGE with residual connections, a global attention mechanism, local gating, and graph contrastive learning to capture both edge features and topological information simultaneously. At the multi-scale level, Liu [4] built MGF-GNN, fusing host-level, port-level, and protocol-level graphs inside a hierarchical message-passing framework, while Zhang et al. [3] addressed adversarial evasion through AEDGNN, a semi-supervised GNN formulation that explicitly models inter-flow relationships to resist labelling noise. Parallel work by Ahanger et al. [9] applied graph attention networks to IoT intrusion detection, and Villegas-Ch et al. [10] developed a dynamic graph modelling approach for IoT traffic. All of these methods share a common limitation with respect to Siren: they operate on discrete-time graph snapshots, produce point predictions, and provide no probabilistic guarantee over the *stationary behaviour* of the monitored system under normal conditions.

2.2 Score-Based Generative Models

The score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ was introduced as a tractable surrogate for the intractable log-likelihood by Hyvärinen [11], who proved that minimising the expected squared difference between the model score and the data score—score matching—equals maximum likelihood under mild regularity conditions. Vincent [12] later showed that denoising autoencoders implicitly perform score matching, motivating the denoising score matching (DSM) objective that sidesteps the intractable Hessian. Building on this, Song and Ermon [13] proposed NCSN, training a noise-conditional score network across a ladder of noise levels and using Langevin dynamics at inference; this multi-scale strategy dramatically improves score estimation quality in high-dimensional settings. Song et al. [5] then unified diffusion-based and score-based generative models under a continuous-time SDE framework, in which the forward process is a prescribed Itô SDE that injects noise and the learned reverse drift is parameterised by the score network. Siren borrows the multi-scale DSM objective and the score-SDE connection from these works, but redirects them toward a fundamentally different objective: rather than generation, we exploit the learned score of the *stationary distribution* of a graph-coupled SDE as a structured anomaly signal.

2.3 Continuous-Time Models on Graphs

Static GNNs [14, 15] capture structural relationships at a single snapshot but are inherently unable to model the continuous temporal evolution of network traffic. Temporal graph networks (TGN) [16] extended GNNs to dynamic graphs through memory modules and temporal attention, enabling event-level updates; however, TGN operates in discrete event time and does not posit a generative stochastic process over the joint graph state. Graph neural diffusion (GRAND) [17] re-interpreted message passing as the discretisation of a *deterministic* heat-diffusion partial differential equation on the graph, providing improved over-smoothing control and depth generalisation; yet the deterministic ODE formulation cannot model the inherent stochasticity of real network traffic. Closest in spirit to our approach, Kidger et al. [18] demonstrated that neural stochastic differential equations can be trained end-to-end via adjoint-based backpropagation through the Itô integral, establishing the computational feasibility of learned stochastic dynamics on sequences. Siren extends this programme to *coupled* systems of Itô SDEs on graphs, where the graph Laplacian enters the drift, the diffusion is learned per-node, and the stationary distribution of the resulting process provides the detection signal—an objective and architecture that is absent from all prior graph–SDE work.

2.4 Stein Operators and Graph Anomaly Detection

The Stein operator $\mathcal{A}_p\, g(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})^\top g(\mathbf{x}) + \nabla_{\mathbf{x}} \cdot g(\mathbf{x})$ satisfies $\mathbb{E}_p[\mathcal{A}_p\, g] = 0$ for all smooth test functions g under mild regularity conditions. Liu et al. [19] exploited this Stein identity to construct the kernelised Stein discrepancy (KSD), a tractable goodness-of-fit measure that does not require normalisation of the reference density—precisely the property that makes it applicable when $p = \pi^*$ is known only through its score. In the graph anomaly detection literature, Cai et al. [20] proposed StrGNN, a structural-temporal GNN that detects anomalous edges by mining unusual temporal subgraph structures; applied to enterprise security, it achieved zero false negatives on benchmark datasets. Most recently, Bai et al. [21] introduced CRC-SGAD, applying conformal risk control to supervised graph anomaly detection with formal bounds on both the false-negative rate and the false-positive rate via a spectral GNN calibrator. CRC-SGAD is the closest existing work to SIREN in combining GNN-based detection with statistical guarantees for a security application; however, it relies on conformal prediction—which requires labelled calibration data—and does not posit or exploit a stochastic process model for normal traffic. SIREN addresses this gap by grounding the anomaly signal in the Stein identity of the *learned* stationary distribution of a graph-coupled Itô SDE, requiring no attack labels at training or calibration time.

2.5 Distributionally Robust Learning on Graphs

Distributionally robust optimisation (DRO) hardens models against distributional shift by minimising the worst-case expected loss within a Wasserstein ball around the empirical distribution [22]. Sadeghi et al. [23] applied this principle to graph-structured data, showing that a Wasserstein-ball DRO objective for GNN semi-supervised classification admits a strong-dual reformulation as regularised empirical risk minimisation. Wang et al. [24] extended graph DRO to collaborative filtering, reinterpreting the GNN as a graph-smoothing regulariser to make the DRO objective tractable at scale. These methods improve worst-case robustness *at training time* for supervised tasks but leave inference uncalibrated and provide no anomaly signal grounded in a generative process model. SIREN takes a complementary approach: rather than robust supervised learning, it learns a continuous-time generative model of normal behaviour and uses deviations from that model's stationary-distribution score as the detection criterion—an unsupervised, process-model-based alternative that requires no attack labels.

3 Methodology

This section details the architecture and formalisms of SIREN. We begin by defining the notation and problem setup, followed by the mechanism for constructing dynamic flow graphs. We then present the core graph-coupled Itō SDE model, the score network architecture, and finally outline the training and inference algorithms.

3.1 Notation and Problem Setup

Let a network at any given time window be represented by a graph $G = (V, E, A)$, where V is the set of hosts (IP addresses), E is the set of observed connections between these hosts, and $A \in \mathbb{R}^{n\times n}$ is the weighted adjacency matrix, with $n = |V|$ denoting the number of active nodes.

For each node $i \in V$, we extract a feature vector $X_i(t) \in \mathbb{R}^d$ summarising its traffic behaviour at continuous time t. The joint state of the network is thus represented by the matrix $X(t) \in \mathbb{R}^{n\times d}$. We define the graph Laplacian as $L = D - A$, where D is the diagonal degree matrix. Let $f_\theta : \mathbb{R}^d \to \mathbb{R}^d$ be a learned neural drift function with parameters θ, and let $\sigma_\phi : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ be a learned diffusion matrix network with parameters ϕ.

Our objective is to model the evolution of $X(t)$ under benign conditions. We assume that benign traffic follows a stationary distribution π^*. By estimating the score function $s^*(x) = \nabla_x \log \pi^*(x)$ using a neural approximation $s_{\theta\phi}(x,t)$, we can monitor live traffic $X(t)$. We flag a node i as anomalous if its empirical trajectory deviates significantly from the model, a deviation we quantify via a *Stein residual* $R_i(t)$.

3.2 Graph Construction and Feature Extraction

SIREN operates on temporal windows of size Δ. Within each window ending at time t, we observe a set of network flows.

Node and Edge Definition: Each unique endpoint (source or destination IP) observed in the current window forms a node in V. We construct a directed edge from host i to host j if at least $k_{\min}$ flows are observed from i to j within the window. To account for varying flow volumes, we assign a logarithmically scaled edge weight:

$$A_{ij} = \frac{\log(1 + \text{flow\_count}(i \to j))}{\log(1 + \text{total\_flows}_i)} \tag{1}$$

For undirected analysis, we symmetrise the adjacency matrix as $A \leftarrow (A + A^T)/2$.

Feature Representation: For each node i, we extract a $d = 12$ dimensional feature vector $X_i(t)$ comprising statistics such as mean flow duration, log-scaled total bytes sent and received, log-scaled packet counts, mean and standard deviation of inter-arrival times, unique source and destination ports, protocol ratios (TCP vs. UDP), and the SYN packet rate. We also append the weighted node degree $\sum_j A_{ij}$ to encapsulate local centrality. These features are Z-score normalised using parameters derived exclusively from the benign training set.

If a node exhibits no traffic in the current window, we apply a zero-order hold, carrying forward its last observed state.

3.3 The Graph-Coupled Itō SDE Model

We model the temporal evolution of the node state $X_i(t)$ as an Itō stochastic differential equation:

$$dX_i(t) = \Big[f_\theta(X_i(t)) + \gamma \sum_{j\in\mathcal{N}(i)} A_{ij}(X_j(t) - X_i(t))\Big]dt + \sigma_\phi(X_i(t))dW_i(t) \tag{2}$$

where $W_i(t)$ is a d-dimensional standard Wiener process.

This formulation consists of three components:

1. **Autonomous Drift (f_θ):** Captures the inherent, host-specific deterministic dynamics.
2. **Topological Mean-Reversion:** The graph Laplacian term scaled by $\gamma \geq 0$ enforces a prior that connected nodes exert a regularising influence on each other, constraining lateral deviation.

3. **Stochastic Diffusion (σ_ϕ):** Captures the inherent burstiness and jitter in network traffic. σ_ϕ outputs a Cholesky factor to guarantee a positive-definite covariance structure.

For discrete-time observation intervals δt, we simulate the trajectory using the Euler-Maruyama scheme:

$$X_i(t+\delta t) = X_i(t) + \Big[f_\theta(X_i(t)) + \gamma \sum_j A_{ij}(X_j(t) - X_i(t))\Big]\delta t + \sigma_\phi(X_i(t))\sqrt{\delta t}\epsilon_i \quad (3)$$

where $\epsilon_i \sim \mathcal{N}(0, I_d)$.

3.4 Score Network Architecture

To detect deviations from the normal stationary distribution π^*, SIREN employs a score network $s_{\theta\phi}(x, \sigma)$ that approximates $\nabla_x \log \pi^*(x)$ conditioned on a noise level σ.

The backbone is a graph-conditioned Multi-Layer Perceptron (MLP) with residual connections. The input $x \in \mathbb{R}^d$ is concatenated with a normalised local neighbourhood context vector $\sum_j A_{ij}X_j/(\deg_i + \epsilon)$. This combined input is projected into a hidden dimension $H = 256$, followed by four residual blocks. We employ EDM-style parameterisation, adding a learned logarithmic noise embedding before the final layer and scaling the output by $1/\sigma$ to maintain numerical stability across varying noise scales.

3.5 Training and Inference Algorithms

Joint SDE and Score Training: SIREN is trained entirely on benign traffic. We jointly optimise the SDE parameters (θ, ϕ) and the score network by minimising a combined loss $L_{total} = L_{score} + \lambda_{sde}L_{sde}$. The SDE loss L_{sde} is the mean squared error of the one-step Euler-Maruyama prediction over observed temporal pairs. The score matching loss L_{score} employs Multi-Scale Denoising Score Matching. We perturb the observed node states with Gaussian noise at $L = 10$ logarithmically spaced levels $\sigma_l \in [\sigma_{\min}, \sigma_{\max}]$. The network is trained to recover the noise vector, minimising the discrepancy against the tractable Gaussian score target $-\epsilon/\sigma_l$.

Inference via Stein Residuals: At inference time, we compute an empirical score estimate $\hat{s}_{emp}(x)$ for each node using a short-window sliding Kernel Density Estimator (KDE). Utilizing the Silverman rule for bandwidth selection, the KDE provides a fast approximation of the local gradient of the data distribution. We define the *Stein residual* for node i as:

$$R_i(t) = \|s_{\theta\phi}(X_i(t), \sigma_{\min}) - \hat{s}_{emp}(X_i(t))\|_2 \quad (4)$$

To uncover stealthy lateral movement, these local residuals are aggregated across the graph:

$$\tilde{R}_i(t) = R_i(t) + \beta \sum_{j \in \mathcal{N}(i)} A_{ij}R_j(t) \quad (5)$$

where β controls the diffusion of the anomaly signal. Finally, an alert is triggered if $\tilde{R}_i(t)$ exceeds a detection threshold τ^*, which is calibrated offline to ensure a target False Alarm Rate (e.g., 1%) on a validation split.

Declarations

Clinical trial registration: Not applicable. This study is not a clinical trial.

Ethics, consent to participate, and consent to publish: Not applicable.

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